Unit 4 - Class Activity

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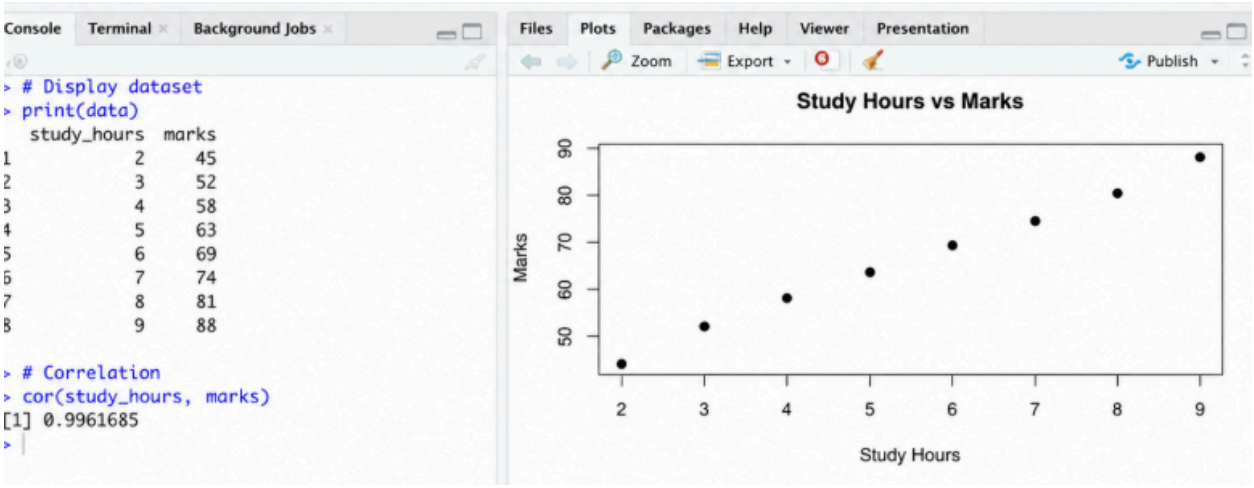
1. Study Hours vs Marks — R

```
# Create dataset
study_hours <- c(2, 3, 4, 5, 6, 7, 8, 9)
marks <- c(45, 52, 58, 63, 69, 74, 81, 88)

data <- data.frame(study_hours, marks)

# Display dataset
print(data)

# Scatterplot
plot(study_hours, marks,
    main = "Study Hours vs Marks",
    xlab = "Study Hours",
    ylab = "Marks",
    pch = 19)

# Correlation
cor(study_hours, marks)
```



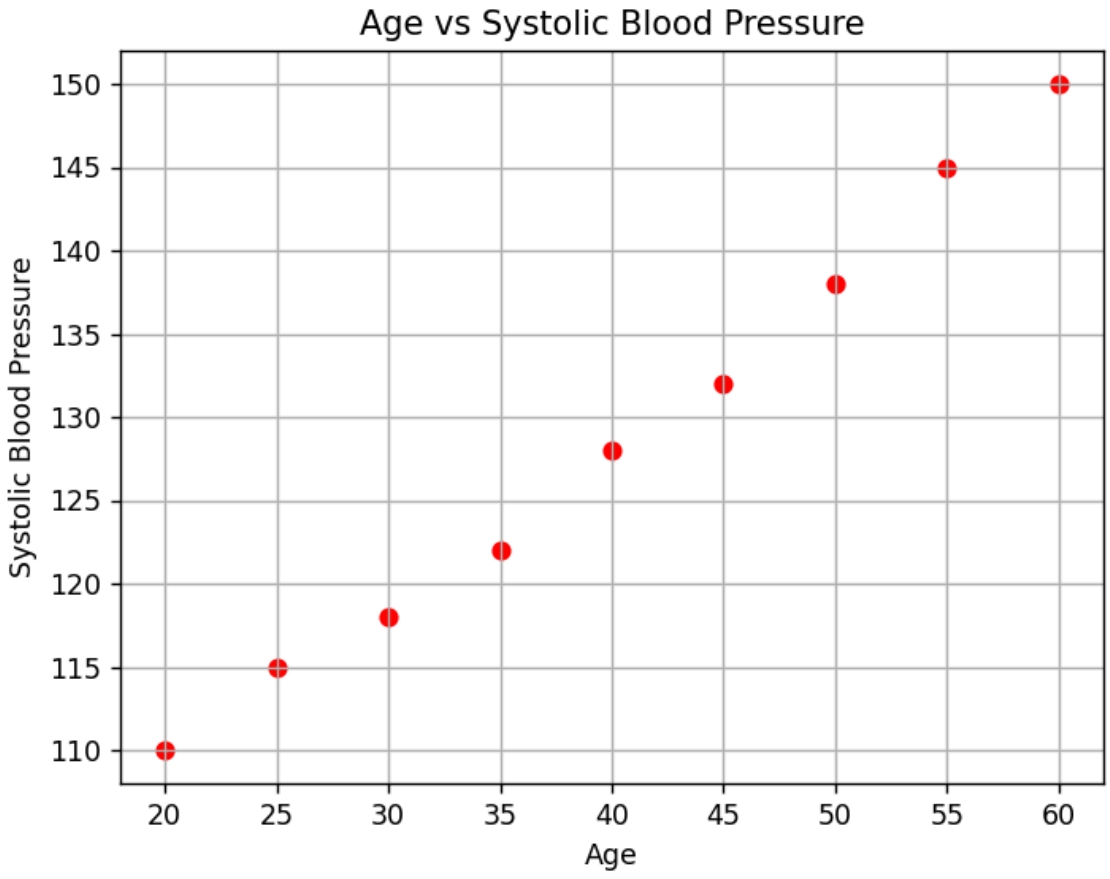
Interpretation:

The scatterplot shows an upward trend. As study hours increase, marks also increase. Therefore, there is a **strong positive correlation** between study hours and marks.

2. Age vs Systolic Blood Pressure — Python

```
import matplotlib.pyplot as plt
```

```
# Dataset
age = [20, 25, 30, 35, 40, 45, 50, 55, 60]
blood_pressure = [110, 115, 118, 122, 128, 132, 138, 145, 150]

# Scatterplot
plt.scatter(age, blood_pressure, color='red')

# Labels and title
plt.xlabel("Age")
plt.ylabel("Systolic Blood Pressure")
plt.title("Age vs Systolic Blood Pressure")

# Gridlines
plt.grid(True)

plt.show()
```

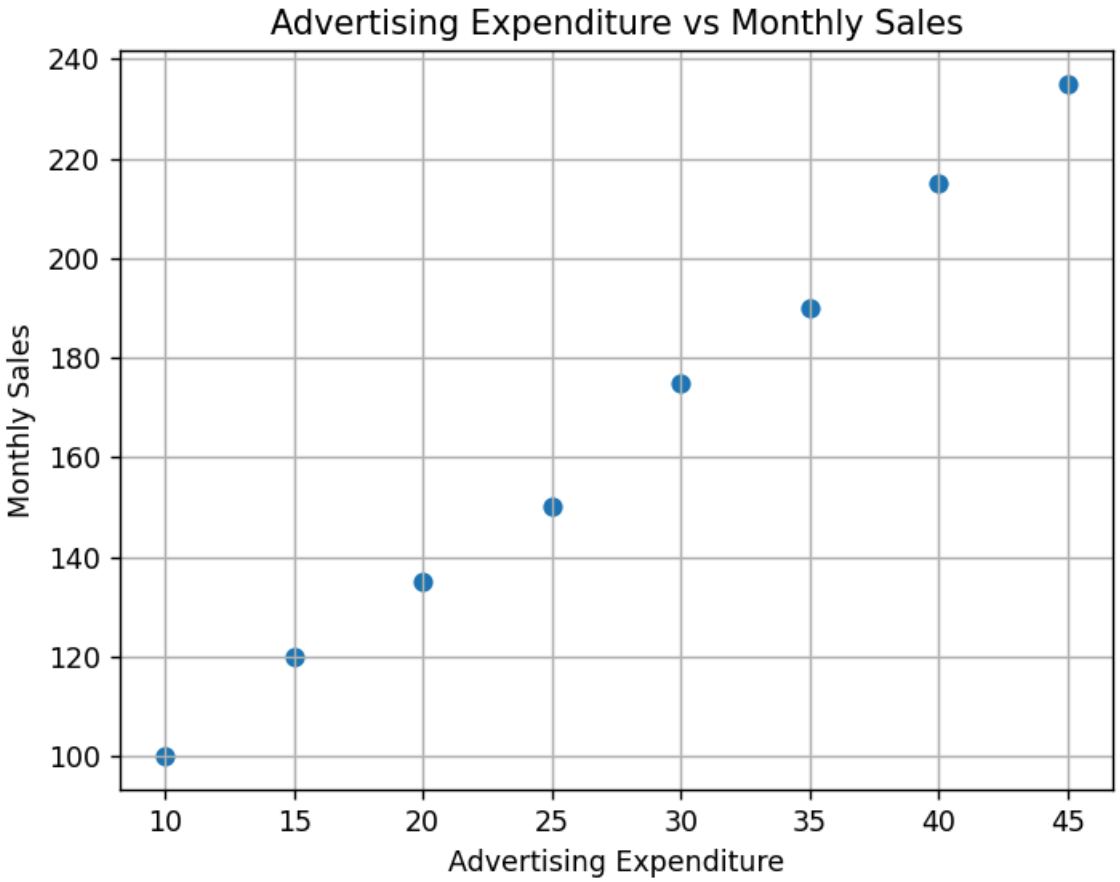


Comment:
The scatterplot shows a **positive trend**. Systolic blood pressure generally increases as age increases.

3. Advertising Expenditure vs Monthly Sales

Python

```
import matplotlib.pyplot as plt

advertising = [10, 15, 20, 25, 30, 35, 40, 45]
sales = [100, 120, 135, 150, 175, 190, 215, 235]

plt.scatter(advertising, sales)

plt.xlabel("Advertising Expenditure")
plt.ylabel("Monthly Sales")
plt.title("Advertising Expenditure vs Monthly Sales")
plt.grid(True)

plt.show()
```



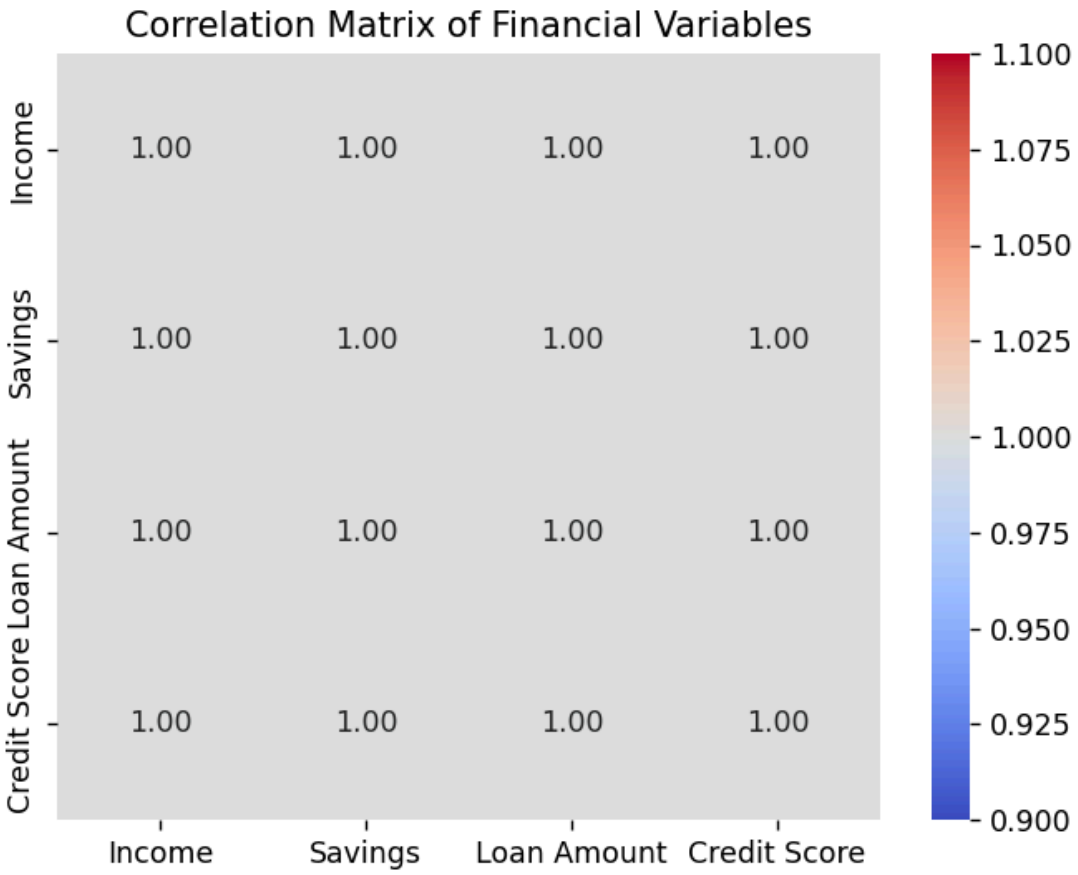
Interpretation:
The scatterplot shows that sales increase as advertising expenditure increases. Hence, there is a **positive correlation** between advertising expenditure and sales.

4. Financial Variables — Correlation Matrix and Correlogram

Python

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Dataset
data = pd.DataFrame({
   'Income': [30, 40, 50, 60, 70, 80, 90, 100],
   'Savings': [5, 10, 15, 20, 25, 30, 35, 40],
   'Loan Amount': [20, 25, 30, 35, 40, 45, 50, 55],
   'Credit Score': [600, 620, 640, 660, 680, 700, 720, 740]
```

```
})

# Correlation matrix
corr = data.corr()

print(corr)

# Correlogram
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Matrix of Financial Variables")
plt.show()
```



	Income	Savings	Loan Amount	Credit Score
Income	1.0	1.0	1.0	1.0
Savings	1.0	1.0	1.0	1.0
Loan Amount	1.0	1.0	1.0	1.0
Credit Score	1.0	1.0	1.0	1.0

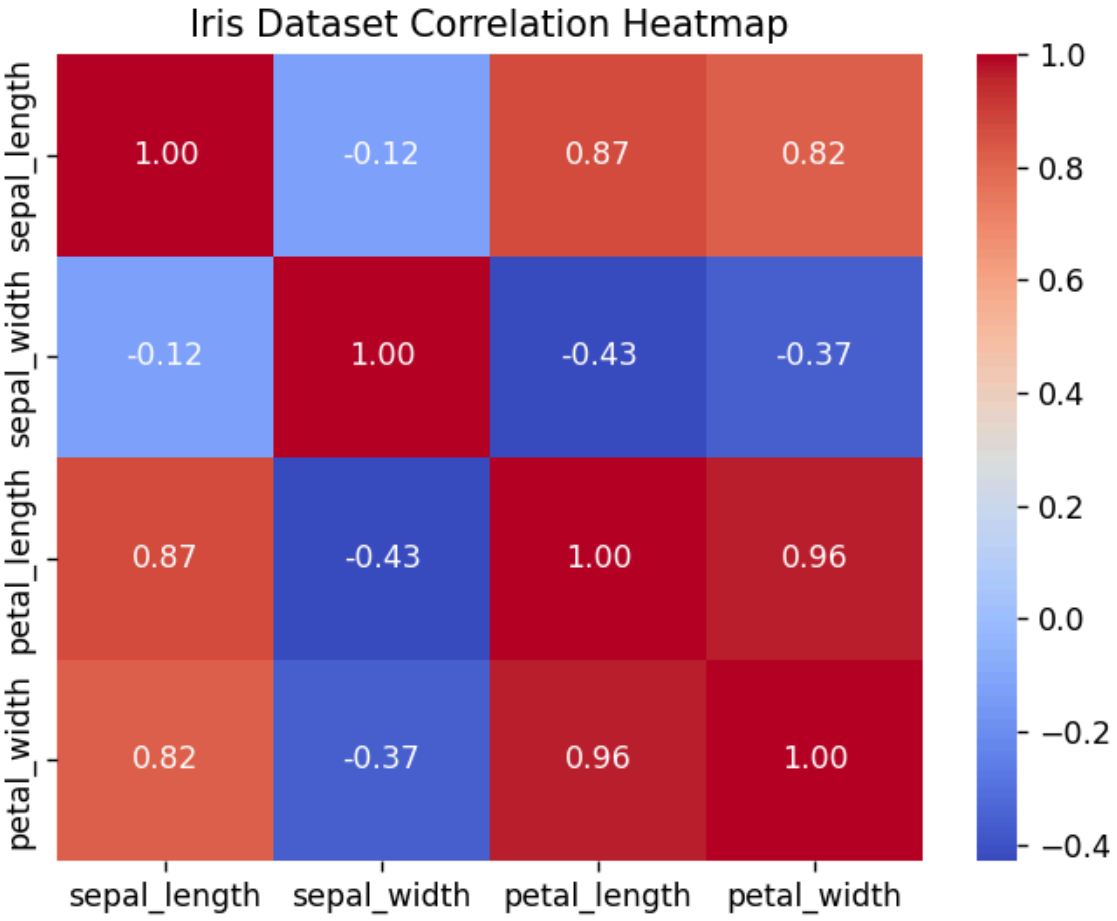
Interpretation:
The correlation matrix shows the relationship between Income, Savings, Loan Amount and Credit Score.

In this example, **Income and Savings** exhibit a very strong positive correlation. The other variables also show positive relationships because of the constructed dataset.

5. Iris Dataset — Correlation Matrix and Heatmap

Python

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load Iris dataset
iris = sns.load_dataset("iris")

# Select numerical columns
data = iris.iloc[:, 0:4]

# Correlation matrix
corr = data.corr()

print(corr)

# Heatmap
sns.heatmap(corr,
        annot=True,
        cmap='coolwarm',
        fmt=".2f")

plt.title("Iris Dataset Correlation Heatmap")
plt.show()
```



	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

Negative correlations:

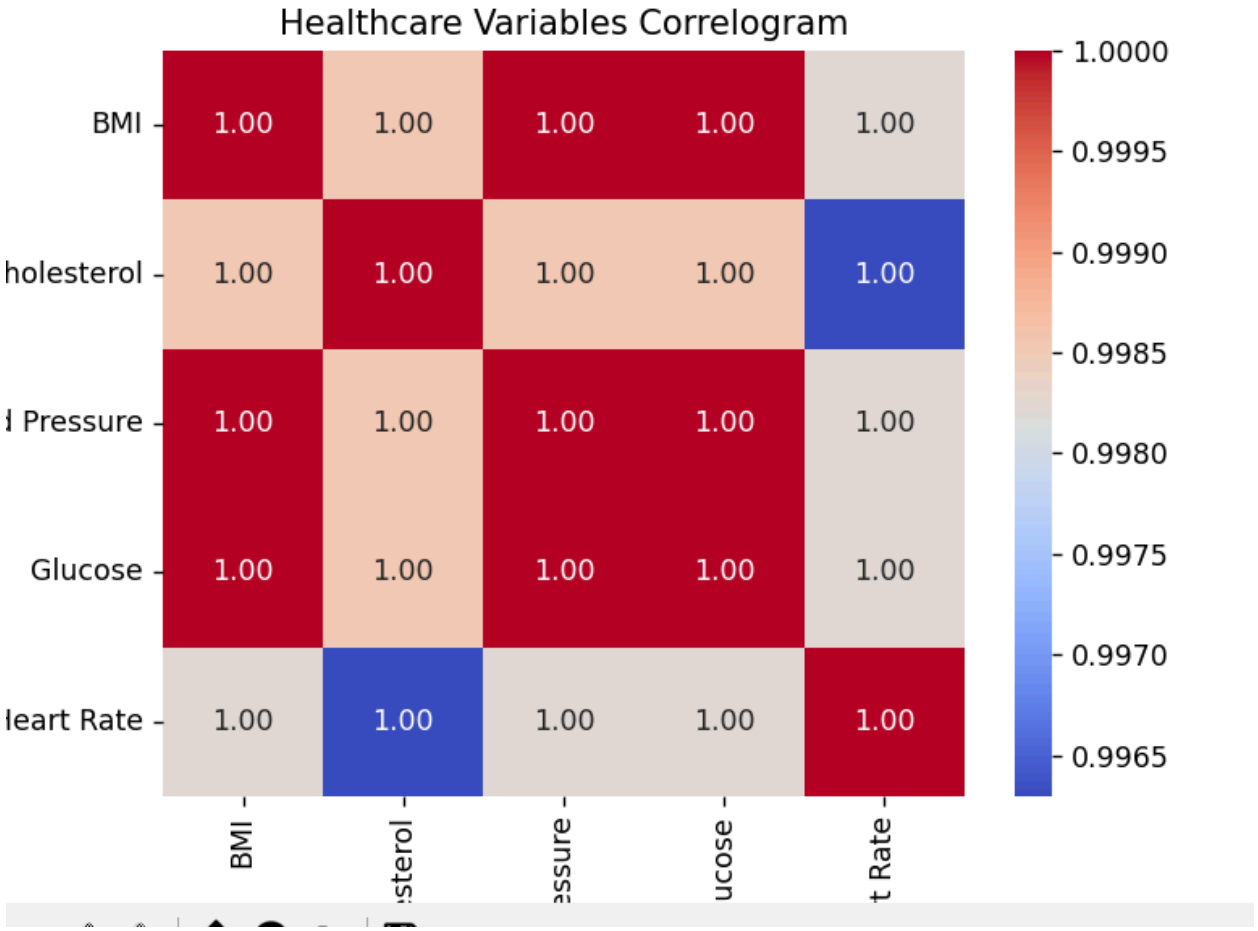
The Iris dataset has a **negative correlation between sepal width and petal length**, and also between **sepal width and petal width**.

6. Healthcare Dataset — Correlogram and Multicollinearity

Python

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data = pd.DataFrame({
  'BMI': [22, 24, 26, 28, 30, 32, 34, 36],
  'Cholesterol': [180, 190, 200, 215, 225, 240, 250, 265],
  'Blood Pressure': [110, 115, 120, 125, 130, 135, 140, 145],
  'Glucose': [80, 90, 100, 110, 120, 130, 140, 150],
  'Heart Rate': [65, 68, 70, 72, 75, 78, 80, 82]
})

# Correlation matrix
corr = data.corr()

print(corr)

# Correlogram
sns.heatmap(corr,
        annot=True,
        cmap='coolwarm',
        fmt=".2f")

plt.title("Healthcare Variables Correlogram")
plt.show()
```



	BMI	Cholesterol	Blood Pressure	Glucose	Heart Rate
BMI	1.000000	0.998552	1.000000	1.000000	0.998214
Cholesterol	0.998552	1.000000	0.998552	0.998552	0.996296
Blood Pressure	1.000000	0.998552	1.000000	1.000000	0.998214
Glucose	1.000000	0.998552	1.000000	1.000000	0.998214
Heart Rate	0.998214	0.996296	0.998214	0.998214	1.000000

Interpretation:

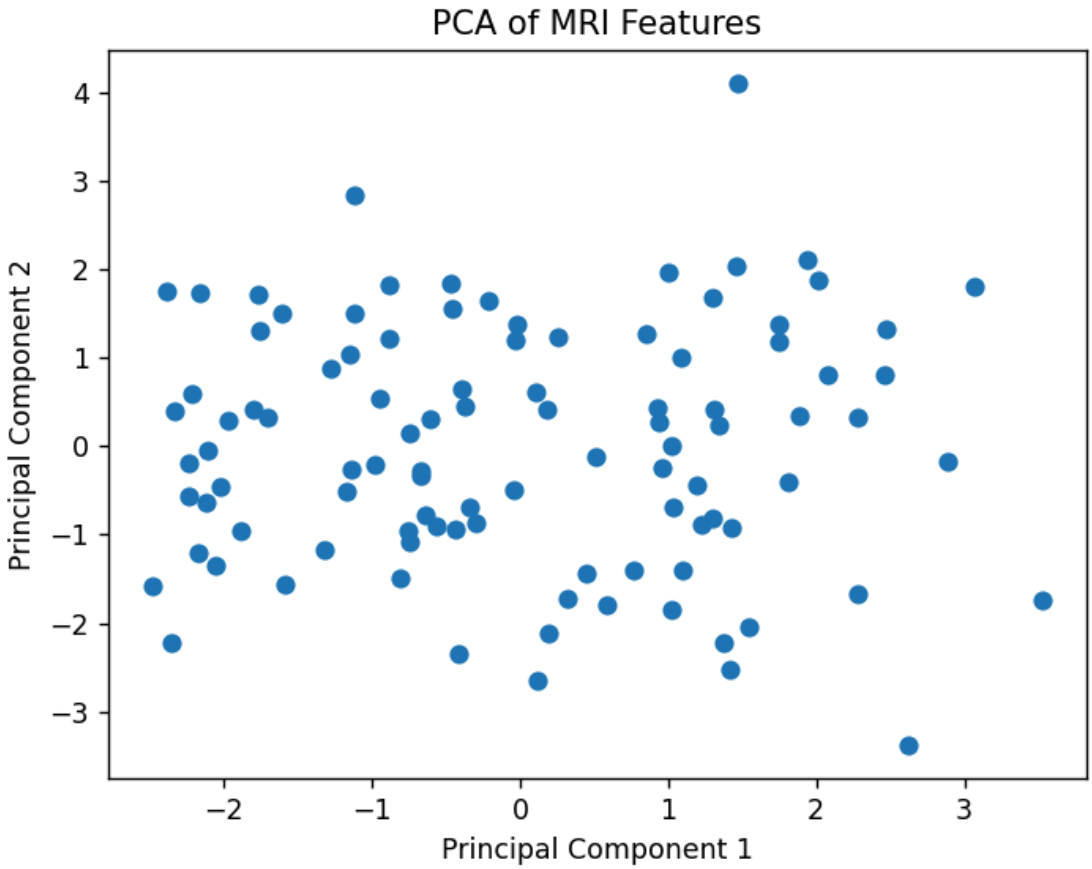
Multicollinearity occurs when two or more independent variables have a **strong correlation** with each other. In the given dataset, several healthcare variables show strong positive correlations, indicating possible multicollinearity.

7. MRI Dataset — PCA to Two Components

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA

# Generate sample MRI dataset
np.random.seed(42)
data = np.random.rand(100, 25)

# Standardize data
scaled_data = StandardScaler().fit_transform(data)

# Apply PCA
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

# Plot transformed data
plt.scatter(pca_data[:, 0], pca_data[:, 1])

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA of MRI Features")

plt.show()

print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)
```




```
Explained Variance Ratio:
[0.09073869 0.07534197]
```

Interpretation:
PCA reduces the original **25 features to 2 principal components** while retaining as much information as possible.

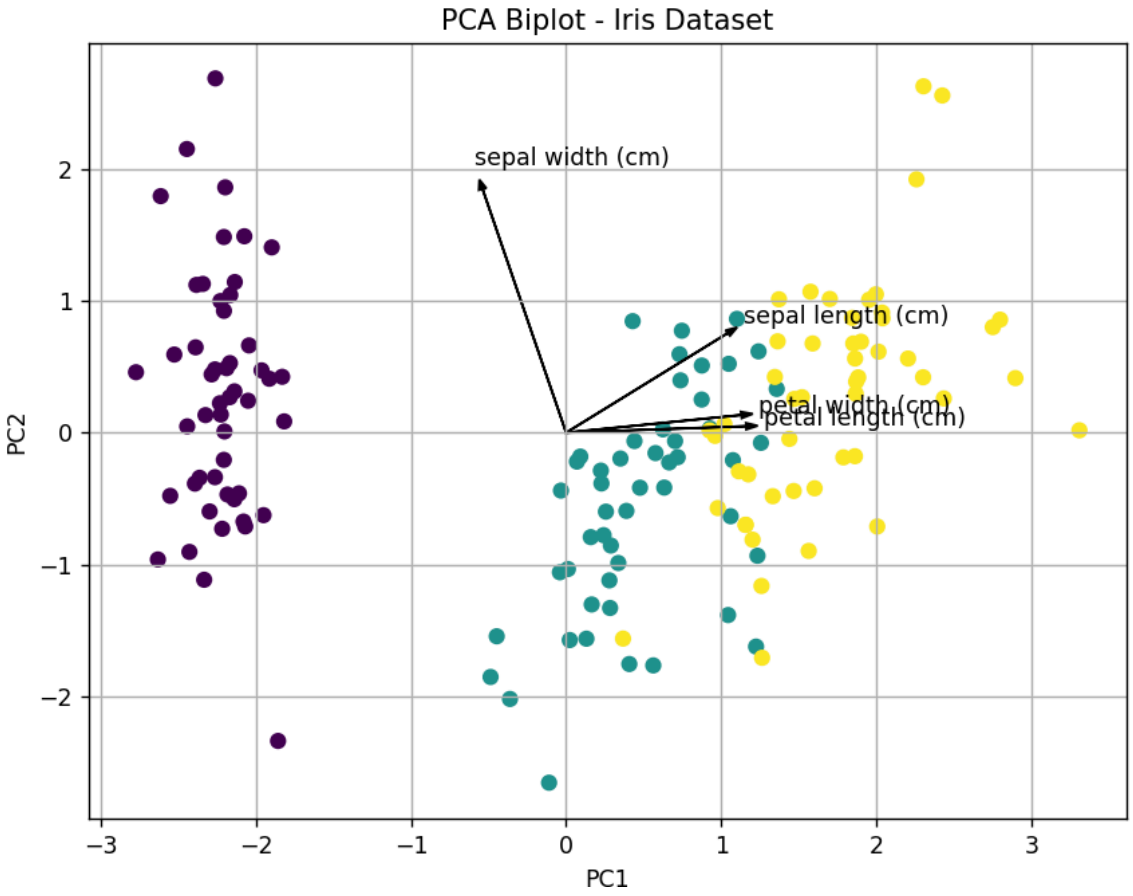
8. Iris Dataset — PCA, Summary and Biplot

Python

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Standardize
X_scaled = StandardScaler().fit_transform(X)

# PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Summary
print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)

print("\nTotal Variance Explained:")
print(sum(pca.explained_variance_ratio_))

# Biplot
plt.figure(figsize=(8, 6))

plt.scatter(X_pca[:, 0],
          X_pca[:, 1],
          c=y)

# Feature vectors
features = iris.feature_names

for i, feature in enumerate(features):
  plt.arrow(0, 0,
            pca.components_[0, i] * 2,
            pca.components_[1, i] * 2,
            color='black',
            head_width=0.05)

  plt.text(pca.components_[0, i] * 2.2,
           pca.components_[1, i] * 2.2,
           feature)
```

```
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA Biplot - Iris Dataset")
plt.grid(True)
plt.show()
```




```
Explained Variance Ratio:
[0.72962445 0.22850762]

Total Variance Explained:
0.9581320720000166
```

Explanation:

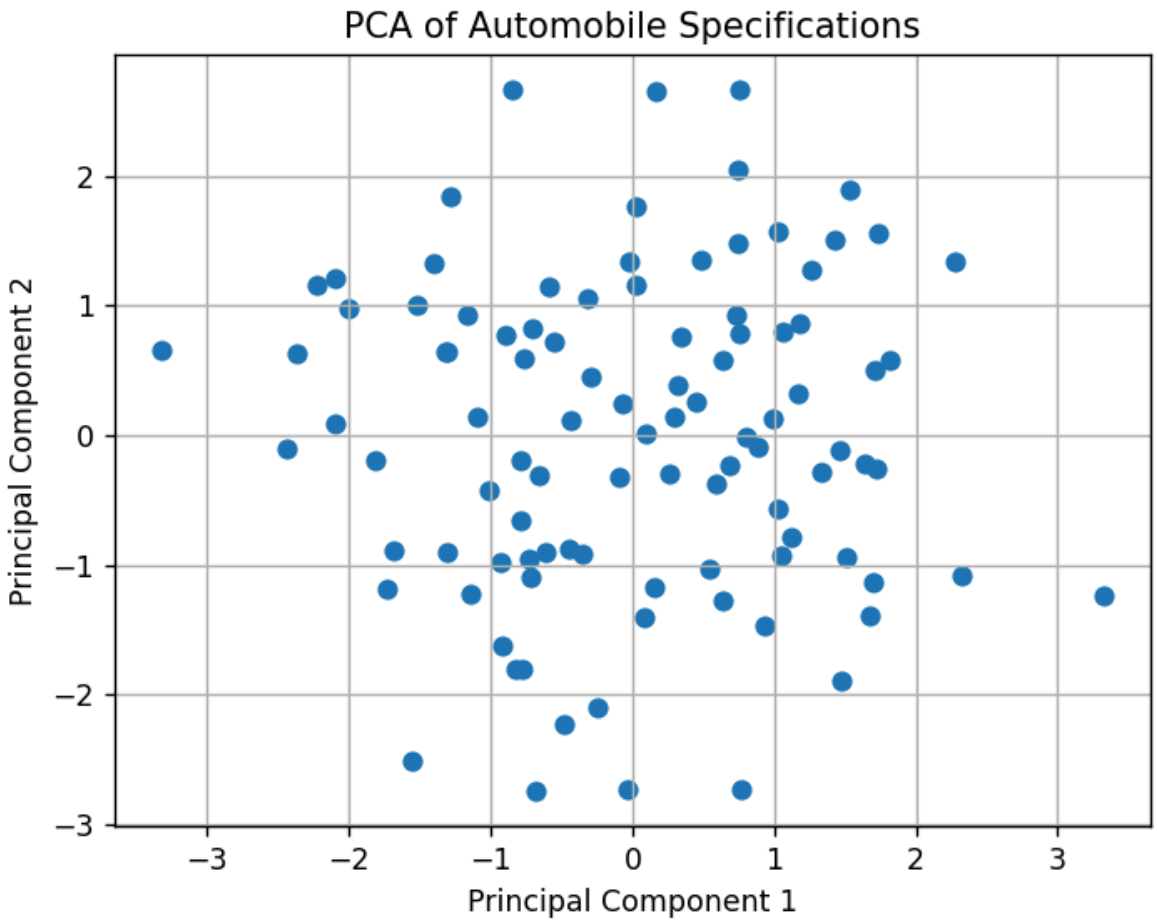
Principal Component 1 (PC1) explains the highest variance because it has the largest explained variance ratio. For the standardized Iris dataset, PC1 explains approximately **73% of the variance**, while PC2 explains approximately **23%**.

Together, PC1 and PC2 explain approximately **96% of the total variance**.

9. Automobile Dataset — PCA

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Generate sample automobile dataset
np.random.seed(10)
data = np.random.rand(100, 15)

# Standardize
scaled_data = StandardScaler().fit_transform(data)

# PCA
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

# Visualization
plt.scatter(pca_data[:, 0], pca_data[:, 1])

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA of Automobile Specifications")

plt.grid(True)
plt.show()

print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)
```



Interpretation:

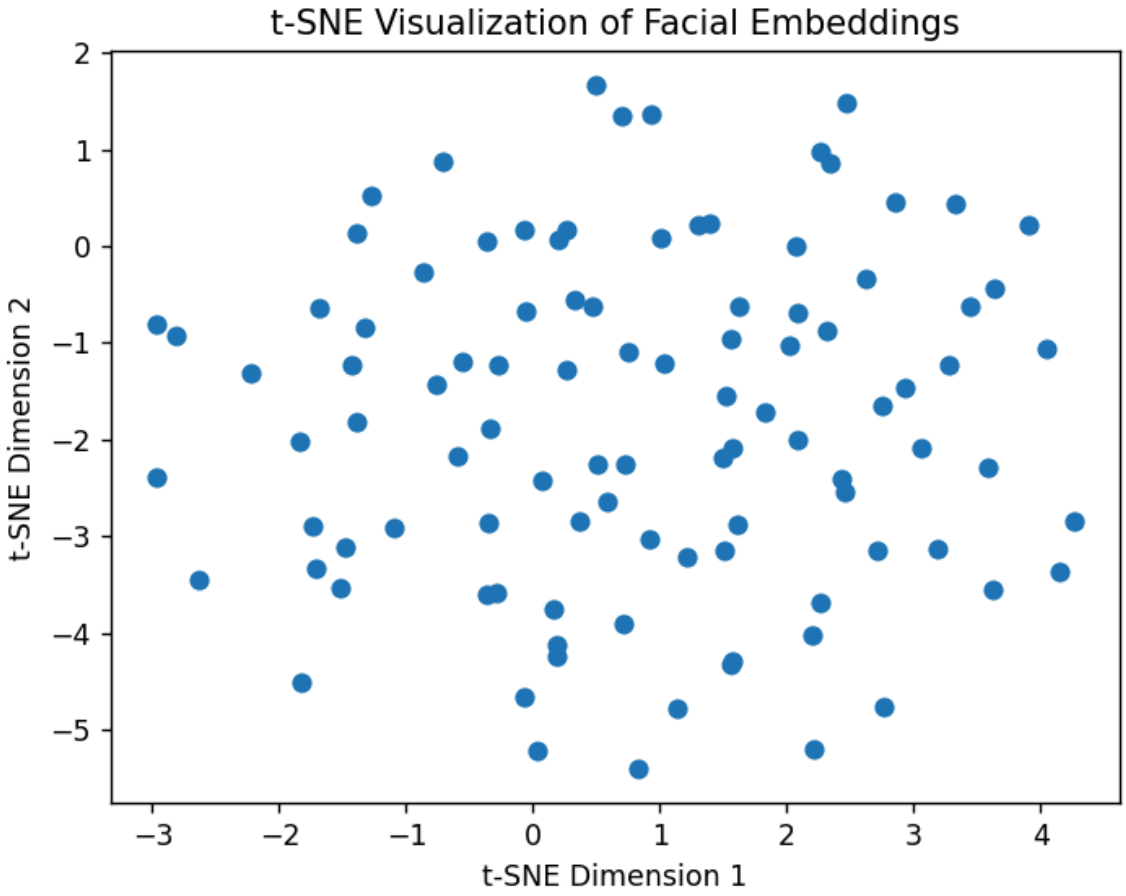
The original **15 automobile specifications** are reduced to two principal components. The scatterplot provides a two-dimensional representation of the automobile data.

10. Facial Recognition — t-SNE

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler

# Generate sample facial embeddings
np.random.seed(42)
data = np.random.rand(100, 128)
```

```
# Standardize
data_scaled = StandardScaler().fit_transform(data)

# Apply t-SNE
tsne = TSNE(n_components=2,
        random_state=42,
        perplexity=30)

data_tsne = tsne.fit_transform(data_scaled)

# Plot
plt.scatter(data_tsne[:, 0], data_tsne[:, 1])

plt.xlabel("t-SNE Dimension 1")
plt.ylabel("t-SNE Dimension 2")
plt.title("t-SNE Visualization of Facial Embeddings")

plt.show()
```



Why t-SNE instead of PCA?

t-SNE is preferred for visualization because it is particularly effective at preserving **local neighborhood relationships** and revealing clusters in high-dimensional data.

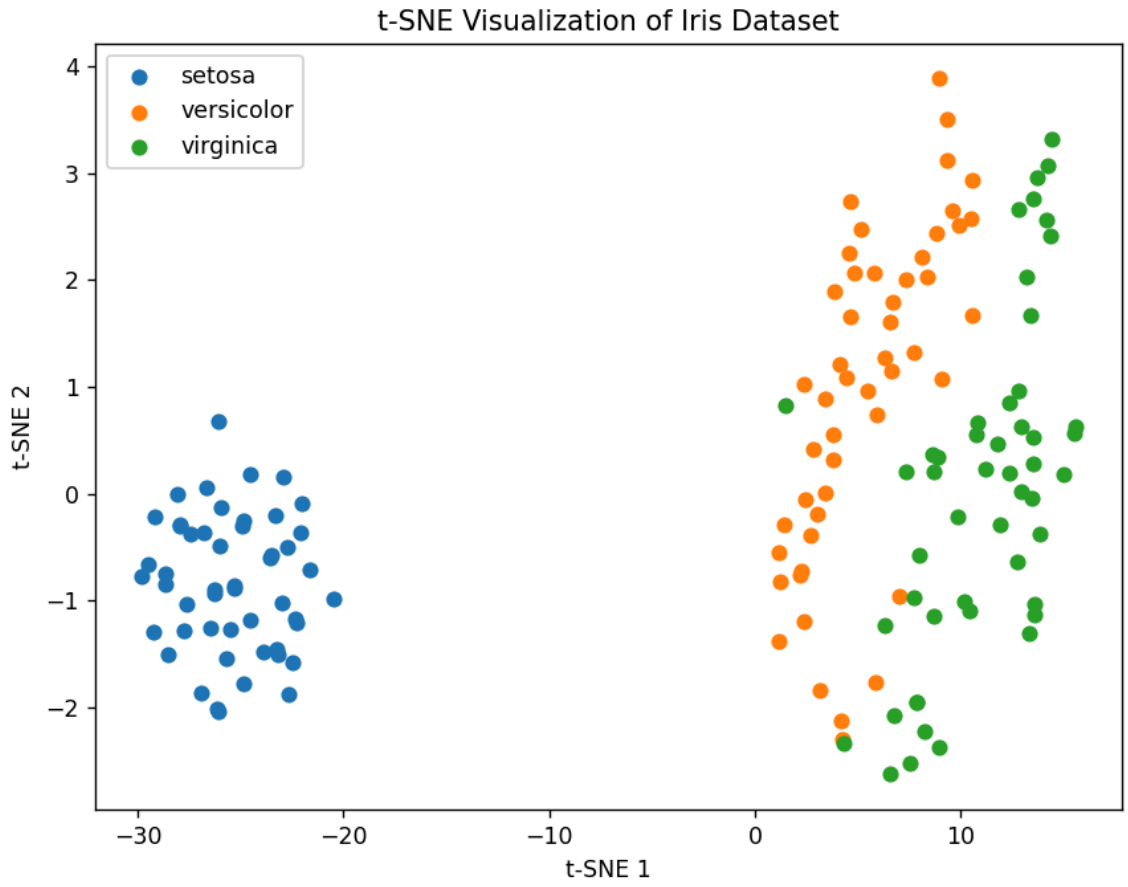
PCA is a **linear dimensionality reduction technique**, whereas t-SNE can capture **non-linear relationships**.

11. Iris Dataset — t-SNE

Python

```
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler
```

```
# Load dataset
iris = sns.load_dataset("iris")

X = iris.iloc[:, 0:4]
y = iris["species"]

# Standardize
X_scaled = StandardScaler().fit_transform(X)

# t-SNE
tsne = TSNE(n_components=2,
        random_state=42,
        perplexity=30)

X_tsne = tsne.fit_transform(X_scaled)

# Plot
plt.figure(figsize=(8, 6))

for species in y.unique():
  index = y == species
  plt.scatter(X_tsne[index, 0],
            X_tsne[index, 1],
            label=species)

plt.xlabel("t-SNE 1")
plt.ylabel("t-SNE 2")
plt.title("t-SNE Visualization of Iris Dataset")
plt.legend()
plt.show()
```



Interpretation:

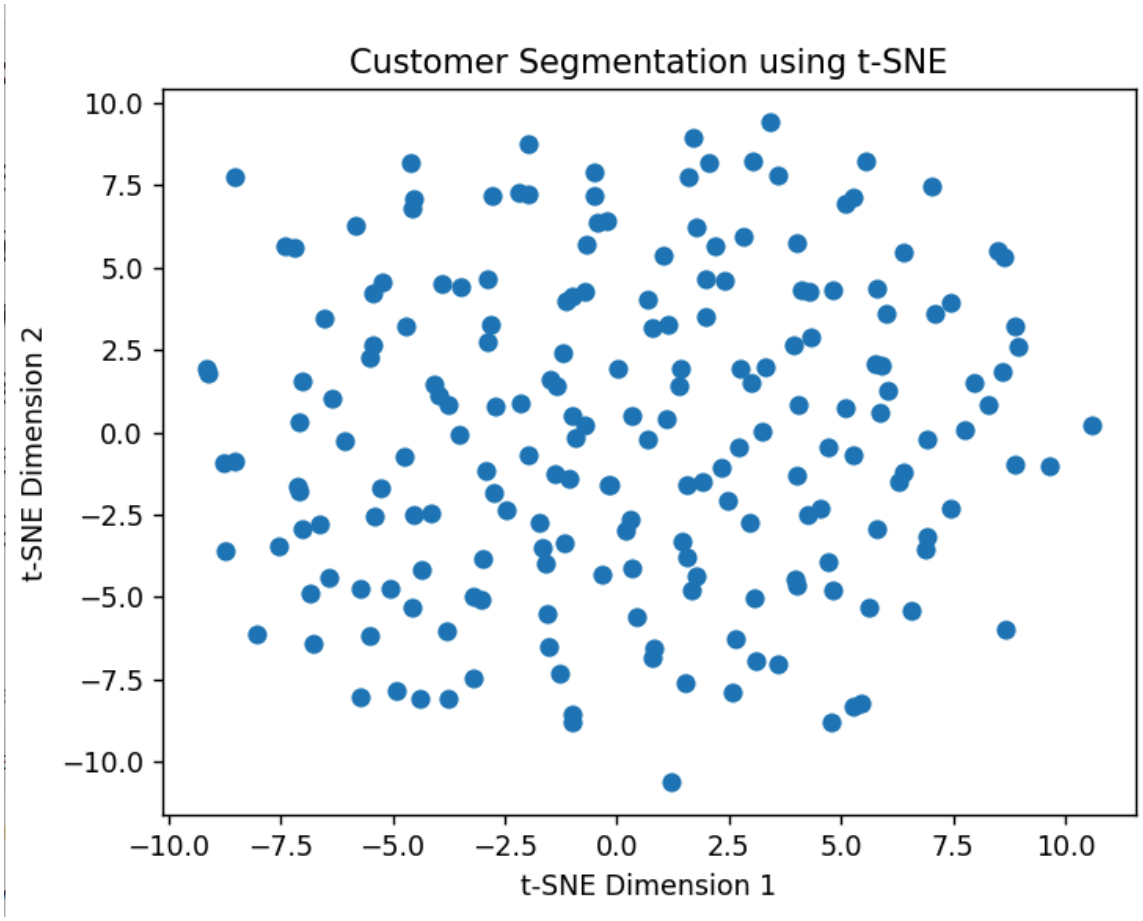
The t-SNE plot generally forms separate groups corresponding to the three Iris species. **Setosa** is usually clearly separated, while **versicolor and virginica** may show some overlap.

12. Customer Segmentation — t-SNE

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler

# Generate sample customer data
np.random.seed(42)
data = np.random.rand(200, 40)
```

```
# Standardize
data_scaled = StandardScaler().fit_transform(data)

# Apply t-SNE
tsne = TSNE(n_components=2,
            random_state=42,
            perplexity=30)

result = tsne.fit_transform(data_scaled)

# Plot
plt.scatter(result[:, 0], result[:, 1])

plt.xlabel("t-SNE Dimension 1")
plt.ylabel("t-SNE Dimension 2")
plt.title("Customer Segmentation using t-SNE")

plt.show()
```



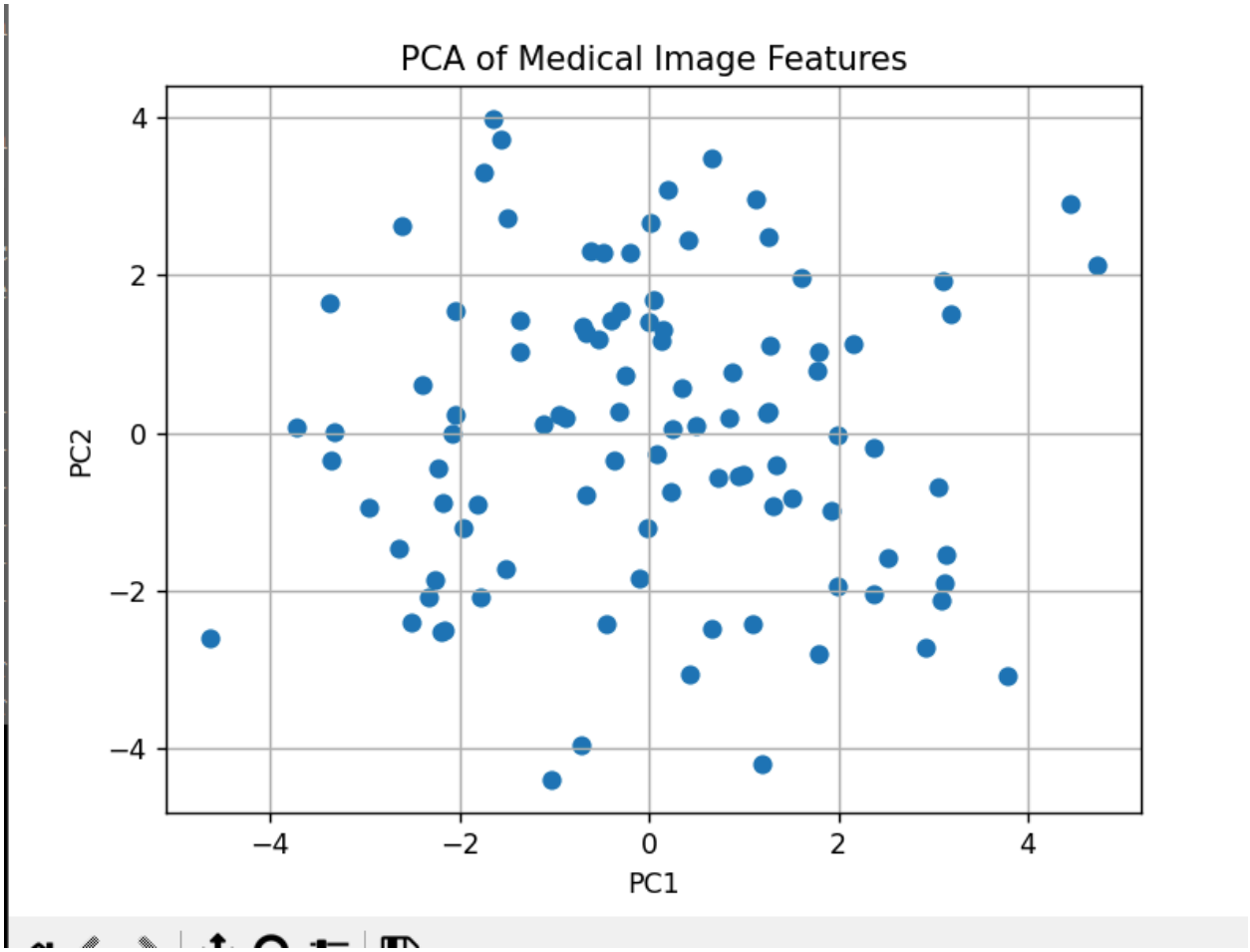
Discussion:
If clearly separated groups appear in the plot, distinct customer clusters exist. If the points are highly mixed, there is no strong evidence of distinct clusters. Since t-SNE is mainly a visualization technique, clustering should ideally be confirmed using methods such as **K-Means or DBSCAN**.

13. Medical Image Dataset — UMAP vs PCA

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import umap

# Generate sample medical image data
np.random.seed(42)
data = np.random.rand(100, 100)

# Standardize
scaled_data = StandardScaler().fit_transform(data)

# UMAP
reducer = umap.UMAP(n_components=2, random_state=42)
umap_data = reducer.fit_transform(scaled_data)

# Plot UMAP
plt.scatter(umap_data[:, 0], umap_data[:, 1])

plt.xlabel("UMAP 1")
plt.ylabel("UMAP 2")
plt.title("UMAP of Medical Image Features")

plt.show()

# PCA for comparison
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

plt.scatter(pca_data[:, 0], pca_data[:, 1])
```

```
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA of Medical Image Features")

plt.show()
```



Comparison

PCA	UMAP
Linear technique	Non-linear technique
Preserves global variance	Preserves local structure well
Faster and simpler	Can reveal complex clusters
Good for dimensionality reduction	Good for visualization and cluster structure

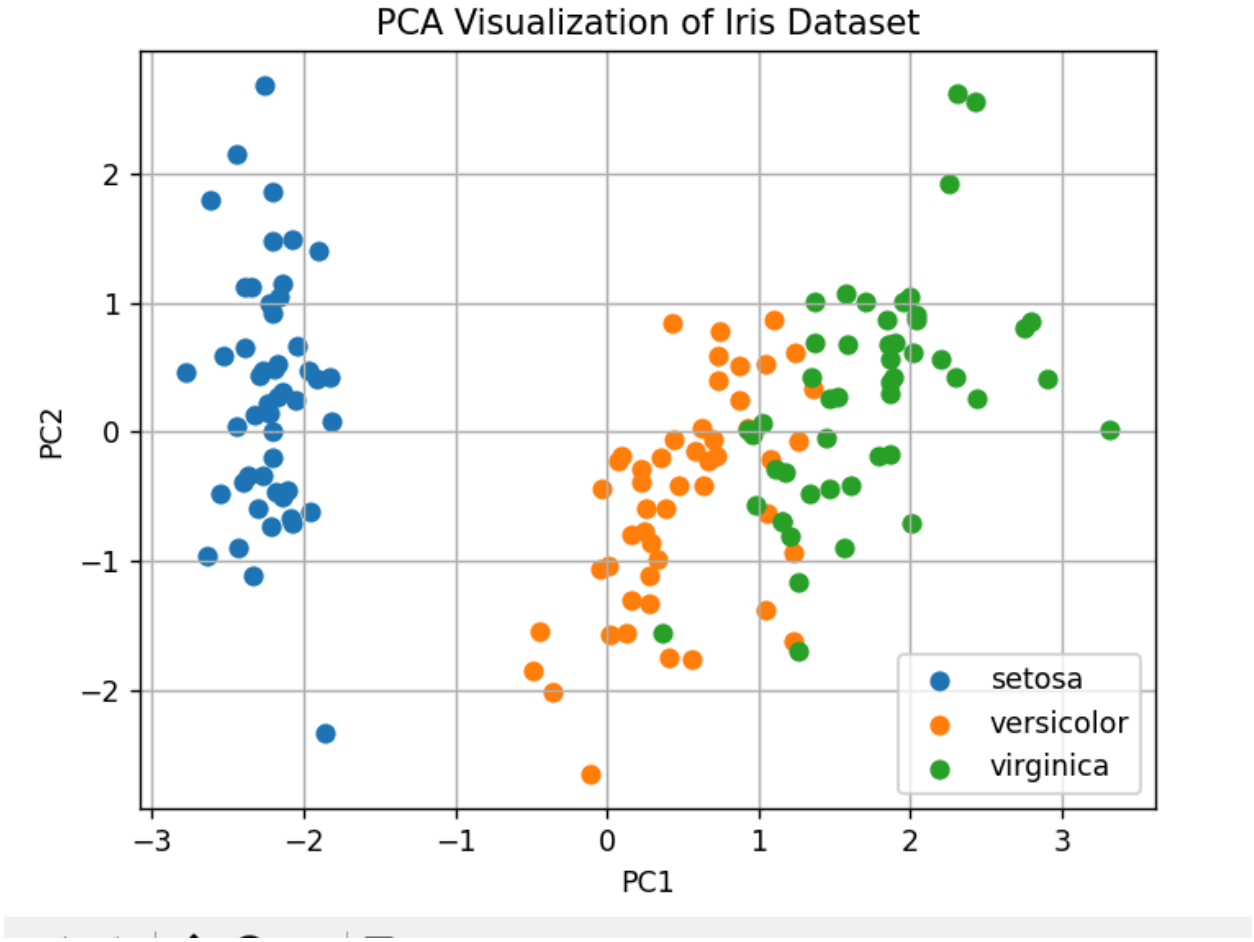
Conclusion:
UMAP can reveal non-linear structures and local clusters that may not be visible using PCA.

14. Iris Dataset — UMAP

Python

```
import seaborn as sns
import matplotlib.pyplot as plt
import umap
from sklearn.preprocessing import StandardScaler

# Load Iris dataset
iris = sns.load_dataset("iris")

X = iris.iloc[:, 0:4]
y = iris["species"]

# Standardize
X_scaled = StandardScaler().fit_transform(X)

# Apply UMAP
reducer = umap.UMAP(n_components=2,
                    random_state=42)

X_umap = reducer.fit_transform(X_scaled)

# Plot
plt.figure(figsize=(8, 6))

for species in y.unique():
  index = y == species

  plt.scatter(X_umap[index, 0],
              X_umap[index, 1],
              label=species)

plt.xlabel("UMAP 1")
plt.ylabel("UMAP 2")
plt.title("UMAP Visualization of Iris Dataset")
plt.legend()
```

plt.show()



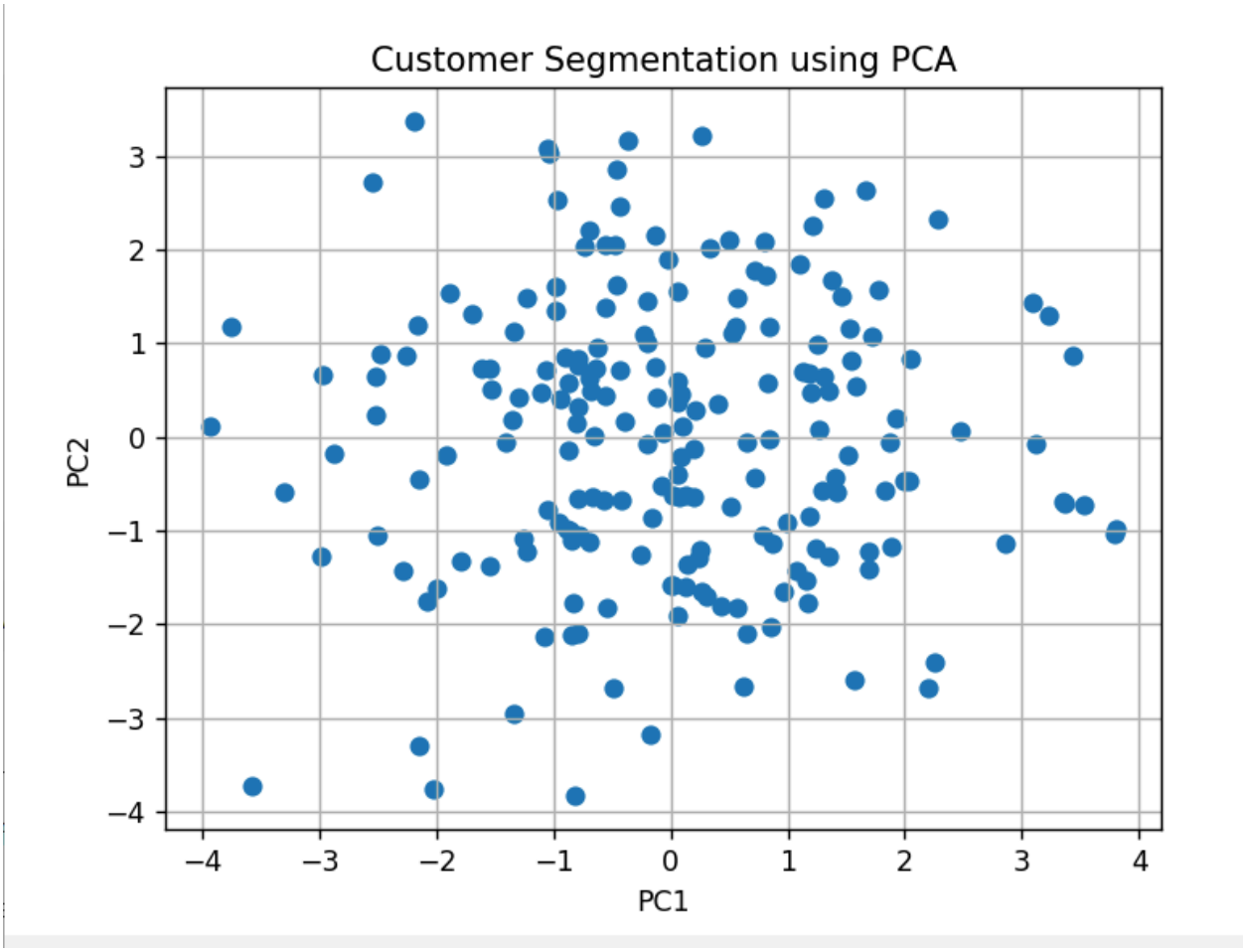
Interpretation:
UMAP generally produces clearly separated groups for the Iris species. **Setosa** is typically well separated, while **versicolor and virginica** can have some overlap.

15. E-Commerce Customer Data — UMAP

Python

```
import numpy as np
import matplotlib.pyplot as plt
import umap
from sklearn.preprocessing import StandardScaler

# Generate sample e-commerce data
np.random.seed(42)
```

```
data = np.random.rand(200, 60)

# Standardize
scaled_data = StandardScaler().fit_transform(data)

# Apply UMAP
reducer = umap.UMAP(n_components=2,
                    random_state=42)

umap_result = reducer.fit_transform(scaled_data)

# Plot
plt.figure(figsize=(8, 6))

plt.scatter(umap_result[:, 0],
            umap_result[:, 1])

plt.xlabel("UMAP Dimension 1")
plt.ylabel("UMAP Dimension 2")
plt.title("Customer Clusters using UMAP")

plt.grid(True)
plt.show()
```



Interpretation:

The UMAP visualization represents the **60-dimensional customer purchasing behavior in two dimensions**. Customers that appear close together have similar behavioral characteristics. Clearly separated groups indicate potential **customer clusters or segments**.